Hanjun Park

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EDUCATION

Ph.D., Industrial and Systems Engineering *08/2025*
Virginia Tech, Blacksburg, VA

Graduate Certificate in Data Analytics
Virginia Tech, Blacksburg, VA

M.S., Industrial Engineering *08/2014*
Seoul National University, Seoul, Korea

B.S., Industrial and Systems Engineering *12/2011*
Georgia Institute of Technology, Atlanta, GA

ACADEMIC APPOINTMENTS

Associated Member, Faculty of Bioengineering *04/2026–Present*
Texas Tech University

Assistant Professor *09/2025–Present*
Texas Tech University

- Department of Industrial, Manufacturing, and Systems Engineering

PUBLICATIONS

Peer-Reviewed Journal Publications (* corresponding-author)

[*J*.9] Santos, F., Son, C.*, **Park, H.,** Xue. C. Yang, J., & Liang, N. (2026). Biomechanical effects of virtual reality headset support and task demands on the neck. *IISE Transactions on Occupational Ergonomics and Human Factors*, 1-19. doi:10.1080/24725838.2026.2680465

[*J*.8] Santos, F., Son, C.*, **Park, H.,** Xue. C, Yang, J., & Liang, N. (2026). Design, fabrication, and testing of an ergonomic intervention for a virtual reality headset using 3D printing. *Ergonomics*, 1-15. doi:10.1080/00140139.2026.2653007

[*J*.7] **Park, H.,** Kim, S., Nussbaum, M.A., & Srinivasan, D.* (2026). Adaptation to a whole-body powered exoskeleton: Human-exoskeleton coordination during load-handling tasks. *Annals of Biomedical Engineering*, 1-15. doi:10.1007/s10439-026-04025-9

[*J*.6] **Park, H.*,** & Nussbaum, M.A. (2025). Muscle synergy analysis during pseudo-static and dynamic overhead tasks with arm-support exoskeletons. *Journal of Biomechanics*, 113135. doi:10.1016/j.jbiomech.2025.113135

[*J*.5] Ransing, V., Park, J., Ye, Y., **Park, H.,** Kim, S., Du, J., & Srinivasan, D.* (2025). Efficacy of virtual reality-based approach for users to understand the potential benefits and limitations of using exoskeletons. *Human Factors*, 67(11), 1136-1151. doi:10.1177/00187208251346627

[*J*.4] **Park, H.,** Noll, A., Kim, S., & Nussbaum, M.A.* (2025). Passive arm-support and back-support exoskeletons have distinct phase-dependent effects on physical demands during cart pushing and pulling: An exploratory study. *Applied Ergonomics*, 126, 104510. doi:10.1016/j.apergo.2025.104510

[*J*.3] **Park, H.,** Kim, S., Nussbaum, M.A., & Srinivasan, D.* (2023). A pilot study investigating motor adaptations when learning to walk with a whole-body powered exoskeleton. *Journal of Electromyography and Kinesiology*, 69, 102755. doi:10.1016/j.jelekin.2023.102755

[*J*.2] **Park, H.,** Kim, S., Nussbaum, M.A., & Srinivasan, D.* (2022). Effects of using a whole-body powered exoskeleton during simulated occupational load-handling tasks: A pilot study. *Applied Ergonomics*, 98, 103589. doi:10.1016/j.apergo.2021.103589

[*J*.1] **Park, H.,** Park, W.*, & Kim, Y. (2014). Manikin families representing obese airline passengers in the U.S. *Journal of Healthcare Engineering*, 5(4), 479-504. doi:10.1260/2040-2295.5.4.479

Manuscripts Under Review

[*M*.6] **Park, H.*,** & Nussbaum, M.A. Muscle activity and joint kinematic adaptations when using passive and active back-support exoskeletons across multiple sessions of repeated symmetric lifting and lowering. *Submitted*

[*M*.5] Jeong, Y., Lee, J.H., **Park, H.**, Choi, W.Y.*, Hwang, D.* How Motion Parameters Shape Perceived Usability of an Autonomous Wheelchair: Age-Related Differences Across Deceleration and Speed Modes. *Submitted*

[*M*.4] Choi, B., Park, J.*, & **Park, H.** Evaluation of passive arm-support exoskeletons in repetitive tasks: physical demands, task performance, and usability effects by task direction. *Submitted*

[*M*.3] **Park, H.*,** & Nussbaum, M.A. Perceptual adaptations to back-support exoskeletons: evidence from measures of workload and discomfort. *In Revision*

[*M*.2] Santos, F., Son, C.*, **Park, H.,** Xue. C. Yang, J., & Liang, N. Ergonomic Design of Head-Neck Supports Using Additive Manufacturing. *In Revision*

[*M*.1] **Park, H.*,** & Nussbaum, M.A. Characterizing adaptation to using exoskeletons: A narrative review and assessment framework for occupational contexts. *In Revision*

Peer-Reviewed Conference Proceedings

[*C*.6] **Park, H.,** Kim, S., Nussbaum, M.A., & Srinivasan, D. (2022). Changes in kinematics and muscle activity when learning to use a whole-body powered exoskeleton for stationary load handling. *Proceedings of the 66th Human Factors and Ergonomics Society Annual Meeting, 66(1),* *273–274.*

[*C*.5] **Park, H.,** Lee, Y., Kim, S., Nussbaum, M.A., & Srinivasan, D. (2021). Gait kinematics when learning to use a whole-body powered exoskeleton. *Proceedings of the 65th Human Factors and Ergonomics Society Annual Meeting, 65(1), 1367–1368.*

[*C*.4] **Park, H.,** Kim, S., Lawton, W., Nussbaum, M.A., & Srinivasan, D. (2020). Effects of using a whole-body powered exoskeleton on physical demands during manual handling. *Proceedings of the 64th Human Factors and Ergonomics Society Annual Meeting, 64(1), 888–889.* **- Won Best Student Paper Award at the Occupational Ergonomics Technical Group**

[*C*.3] Jung, J., **Park, H.,** Hwang, D., Son, M., Beck, D., Park, J., & Park, W. (2014). A review on interaction techniques in virtual environments. *Proceedings of the 2014 International Conference on Industrial Engineering and Operations Management (pp. 1582-1590).*

[*C*.2] Beck, D., Hwang, D., Son, M., Jung, J., **Park, H.,** Park, J., & Park, W. (2013). A Review on Techniques for Evaluating Physical Stresses Associated with the Use of Various Input Devices. *Proceedings of the 2013 Fall Conference of Ergonomics Society of Korea (ESK), Jeju, Korea*.

[*C*.1] Jung, J., **Park, H.,** Hwang, D., Son, M., Beck, D., Park, J., & Park, W. (2013). A Review on Object Manipulation Techniques in Virtual Environments and their Performance Metrics. *Proceedings of 2013 Fall Conference of Ergonomics Society of Korea (ESK), Jeju, Korea*.

FUNDED RESEARCH PROJECTS AND GRANTS

External Research Grants

[*EG*.1] **Biomechanical and Usability Assessment of Occupational Exoskeletons for Nurses in Overweight and Obese Patient Handling**
Sponsor: Heartland Center for Occupational Health & Safety Pilot Project Award (NIOSH ERC)
Total Amount: $19,337
Dates: 2026 – 2027
Role: Principal Investigator
Co-PI: Changwon Son

PRESENTATIONS

Conference Oral Presentations

[*O*.13] **Park, H.,**, & Nussbaum, M.A. (2026). Adaptation to using passive and active back-support exoskeletons during repetitive lifting tasks. *IISE Annual Conference & Expo 2026, Arlington, TX.*

[*O*.12] Santos, F., & Son, C., **Park, H.,** (2026). What Happens to Your Neck when You Wear VR for Long: Effects of Task Demands and Ergonomic Supports on Muscle Activity and Kinematics. *IISE Annual Conference & Expo 2026, Arlington, TX.*

[*O*.11] Santos, F., & Son, C., **Park, H.,** (2026). What Happens to Your Neck when You Wear VR for Long: Effects of Task Demands and Ergonomic Supports on Muscle Activity and Kinematics. *HFES Regional Meeting, College Station, TX*.

[*O*.10] Santos, F., & Son, C., **Park, H.,** (2026). A 3D-printed shoulder support to reduce neck strain during virtual reality device use. *30th Applied Ergonomics Conference, Arlington, TX*.

[*O*.9] Santos, F., **Park, H.,** & Son, C. (2025). A 3D-printed shoulder support to reduce neck strain during prolonged virtual reality (VR) device use: pilot findings. *69th Human Factors and Ergonomics Society Annual Meeting, Chicago, IL*.

[*O*.8] **Park, H.**, Ojelade A., Kim, S., & Nussbaum, M.A. (2024). Effects of using arm-support exoskeletons on muscle coordination during a pseudo-static overhead task: preliminary analysis of muscle synergy. *68th Human Factors and Ergonomics Society Annual Meeting, Phoenix, AZ*. **- Won Best Experimental Paper Award at the Occupational Ergonomics Technical Group.**

[*O*.7] **Park, H.**, Noll, A., Kim, S., & Nussbaum, M.A. (2023). Effects of using passive back- and arm-support exoskeletons for cart pushing and pulling. *67th Human Factors and Ergonomics Society Annual Meeting, Washington, DC*.

[*O*.6] Noll, A., Nussbaum, M.A., Kim, S., & **Park, H.** (2023). Reliability Analysis of Task Performance and Subjective Measures for Assessment of Occupational Exoskeletons. *67th Human Factors and Ergonomics Society Annual Meeting, Washington, DC*.

[*O*.5] **Park, H.**, Kim, S., Nussbaum, M.A., & Srinivasan, D. (2022). Changes in kinematics and muscle activity when learning to use a whole-body powered exoskeleton for stationary load handling. *66th Human Factors and Ergonomics Society Annual Meeting, Baltimore, MD*.

[*O*.4] **Park, H.**, Lee, Y., Kim, S., Nussbaum, M.A., & Srinivasan, D. (2021). Gait kinematics when learning to use a whole-body powered exoskeleton. *65th Human Factors and Ergonomics Society Annual Meeting, Baltimore, MD*.

[*O*.3] **Park, H.**, Kim, S., Lawton, W., Nussbaum, M.A., & Srinivasan, D. (2020). Effects of using a whole-body powered exoskeleton on physical demands during manual handling. *64th Human Factors and Ergonomics Society Annual Meeting*.

[*O*.2] Beck, D., Hwang, D., Son, M., Jung, J., **Park, H.**, Park, J., & Park, W. (2013). A Review on Techniques for Evaluating Physical Stresses Associated with the Use of Various Input Devices. *2013 Fall Conference of Ergonomics Society of Korea (ESK), Jeju, Korea*.

[*O*.1] Jung, J., **Park, H.,** Hwang, D., Son, M., Beck, D., Park, J., & Park, W. (2013). A Review on Object Manipulation Techniques in Virtual Environments and their Performance Metrics. *2013 Fall Conference of Ergonomics Society of Korea (ESK), Jeju, Korea*.

Poster Presentations

[*P*.4] **Park, H.,** & Nussbaum, M.A. Motor Adaptation and Perceived Exertion During Repetitive

Lifting With Passive and Active Occupational Back-Support Exoskeletons. (2026). *10th World Congress of Biomechanics (WCB 2026), Vancouver, Canada, July*.

[*P*.3] Santos, F., Son, C., **Park, H.,** Xue. C, Yang, J., & Liang, N. Design, fabrication, and testing of an ergonomic intervention for virtual reality headset using 3D printing. (2026). *WCOE Research Day, Texas Tech University, Lubbock, TX, February*.

[*P*.2] **Park, H.,** Ojelade, A., Kim, S., & Nussbaum, M. A. (2023). Effects of different arm-support exoskeletons on muscle activation patterns during a pseudo-static overhead task - preliminary analysis of muscle synergy. *37th American Society of Safety Professionals (ASSP) Region VI Professional Development Conference (PDC) in Myrtle Beach, South Carolina, September*. **- Won Student Presentation Award**

[*P*.1] **Park, H.,** Kim, S., Nussbaum, M. A., & Srinivasan, D. (2020). Effects of using a whole-body powered exoskeleton on physical demands during manual material handling. *2020 Annual HFES/INFORMS Poster Competition, Virginia Tech, VA, October*. **- Won Student Poster Presentation Award**

Invited Seminars

[*S*.3] **Park, H.** (2025). Characterizing Human Motor Adaptation and Learning When Using Different Occupational Exoskeletons. *Invited Seminar at the Department of Psychological Sciences, Texas Tech University, October*.

[*S*.2] **Park, H.** (2025). Characterizing Human Motor Adaptation and Learning When Using Different Occupational Exoskeletons. *Invited Seminar at the Department of Industrial, Manufacturing, and Systems Engineering, Texas Tech University, April*.

[*S*.1] **Park, H.** (2025). Characterizing Human Motor Adaptation and Learning When Using Different Occupational Exoskeletons. *Invited Seminar at the Department of Industrial and Systems Engineering, Kennesaw State University, March*.

Invited Panel Talk

[*T*.1] **Park, H.** (2025). From Student to Professor: Exploring Careers in Occupational Ergonomics. Panel Member. *Occupational Ergonomics Technical Group of HFES, September*.

HONORS & AWARDS

- Best Experimental Paper Award, OETG at HFES *2024*
- Best Student Poster Presentation Award, ASSP Regional Conference *2023*
- IISE Future Faculty Fellow (3F) Program Awardee *2023–2024*
- NIOSH Training Fellow, Virginia Tech *2023–2025*
  - Fellowship includes monthly stipend, tuition, and required fees
- ISE Graduate Travel Award ($900), Virginia Tech *2022*
- Best Student Paper Award, OETG at HFES *2020*

- VT HFES Student Chapter Research Competition Award *2020*
- Grado Department of ISE Fellowship, Virginia Tech *2018*
    - Fellowship includes monthly stipend, tuition, and required fees
- Brain Korea 21 Scholarship, Seoul National University *2014*
- Lecture & Research Scholarship, Seoul National University *2013*

TEACHING EXPERIENCE

Instructor of Record

Texas Tech, Dept. of Industrial, Manufacturing, & Systems Engineering

- IE 5304: Biomechanics & Work Physiology (n = 11) *Spring, 2026*

Virginia Tech, Dept. of Industrial and Systems Engineering

- ISE 4624: Work Physiology (n = 36) *Fall, 2023*
- ISE 3614: Human Factors Engineering and Ergonomics (taught online, n = 7) *Summer, 2023*

Guest Lecturer

Virginia Tech, Dept. of Industrial and Systems Engineering

- ISE 3624: Industrial Ergonomics *Spring, 2023*
  "NIOSH Lifting Equation and Controls" (n = 100)
- ISE 4644: Occupational Safety and Hazards Control *Fall, 2018*
  "Review on Hazard Analysis and Prevention" (n = 20)

Graduate Teaching Assistant

Virginia Tech, Dept. of Industrial and Systems Engineering

- ISE 3624: Industrial Ergonomics *Spring, 2019*
- ISE 4624: Work Physiology *Spring, 2019*
- ISE 4644: Occupational Health and Hazards Control *Fall, 2018*

PROFESSIONAL EXPERIENCES

Software Engineer, *Military duty* *10/2014–03/2018*
Kornic Glory, Seoul, Korea

Research Intern *06/2011–08/2011*
LG Electronics, Seoul, Korea

ADVISING & MENTORING

Dissertation/Thesis Committee Member

- Felipe Santos, TTU (PhD, IMSE) (*2025–Present*) [*J*.9], [*J*.8], [*O*.12], [*O*.11], [*O*.10], [*O*.9], [*U*.2]
    Chair: Changwon Son
- Daniel Leibman, TTU (PhD, Psychology) (*2025–Present*)
    Chair: Heesun Choi
- Armina Rahman, TTU (PhD, IMSE) (*2025–Present*)
    Chair: Changwon Son

- Jessica Lillian Dedeaux, TTU (PhD, IMSE) (*2026–Present*)
  Chair: Changwon Son

Research Mentoring

Voluntary Postdoc Position

- Leonardo Wei, Texas Tech (PhD, ISE) (*02/2026–03/2026*)

M.S. Students

- Delaney Sheppard, Texas Tech (BIOE) (*02/2026–Present*)
- Alex Noll, Virginia Tech (ISE) (*08/2022–08/2024*) [*J*.4], [C.8], [C.7]

Undergraduate Students

- Enriquez Bazoalto Kaylee, Virginia Tech (ISE) (*05/2025-08/2025*)

PROFESSIONAL DEVELOPMENT AND SERVICE ACTIVITIES

Professional Development Training

- IISE New Faculty Colloquium - IISE *2026*
- IISE Future Faculty Fellow (3F) Program - IISE *2023-2024*

Service to Profession

- Session Chair, IISE Annual Conference & Expo 2026, Arlington, TX *2026*
- Session Chair, the 69th HFES Annual Meeting, Atlanta, GA *2025*

Archival Journal

- Reviewer, *Ergonomics* *2026–Present*
- Reviewer, *Journal of Biomechanics* *2026–Present*
- Reviewer, *Gait and Posture* *2026–Present*
- Reviewer, *IEEE Transactions on Neural Systems and Rehabilitation Engineering* *2026–Present*
- Reviewer, *IEEE Transactions on Human-Machine Systems* *2025–Present*
- Reviewer, *IISE Transactions on Occupational Ergonomics and Human Factors* *2023–Present*
- Reviewer, *Human Factors and Ergonomics Society Proceedings* *2021–Present*

Service to Institution

Texas Tech Regional Science Fair (K-12) Special Award Judge *2026*
Texas Tech Regional Science Fair (K-12) Ribbon Judge *2026*
Dean's Representative, Dissertation Defense of Yves Valentin
(MS in Human Factors Psychology, TTU Department of Psychological Sciences) *2026*

TTU Graduate Academic Advisor *2025–Present*

VT Chapter of the Human Factors and Ergonomics Society (HFES)

- Officer: *Webmaster* *2020–2022*

National Biomechanics Day

- *Led lab tours and biomechanics activities for local high school students* *2020–2025*

VT Center for the Enhancement of Engineering Diversity (CEED) TechGirls Summer Camp

- *Led hands-on biomechanics activities* *2019*

VT College of Engineering Alumni Outreach Event

- *Led lab tours for alumni* *2019*

PROFESSIONAL AFFILIATIONS

Member, Institute of Industrial and Systems Engineers (IISE) *2023–Present*
Member, Human Factors and Ergonomics Society (HFES) *2020–Present*
Steering Committee member, K-Human Factors and Ergonomics Society (K-HFES) *2021–Present*